Effect of *Plasmodium falciparum* on the Survival of Naturally Infected Afrotropical *Anopheles* (Diptera: Culicidae)

GEOFFREY M. M. CHEGE AND JOHN C. BEIER[1]

Kenya Medical Research Institute and U.S. Army Medical Research Unit, Nairobi, Kenya. Box 401, APO New York 09675

J. Med. Entomol. 27(4): 454–458 (1990)

ABSTRACT The effect of the malarial parasite, *Plasmodium falciparum* Welch, on the daily survival rates and longevity of *Anopheles gambiae* Giles sensu lato and *Anopheles funestus* Giles was determined for wild-caught, naturally infected females from western Kenya. Mosquitoes were collected inside houses and held in cages until death, after which they were assayed for *P. falciparum* circumsporozoite protein by an enzyme-linked immunosorbent assay (ELISA). Survival rates of field populations determined by parity rates were significantly higher than survival rates estimated by regression for *Anopheles* dying in cages. Overall, *An. gambiae* s.l. had a significantly higher daily mortality rate ($\bar{x}$ = 17.5%) than *An. funestus* ($\bar{x}$ = 13.2%). *P. falciparum* ELISA infection rates, which were higher for *An. gambiae* s.l. ($\bar{x}$ = 19.8%; n = 1, 221) than for *An. funestus* ($\bar{x}$ = 11.9%; n = 1, 128), did not increase as a function of time for caged *Anopheles*. For *An. gambiae* s.l., there was a significant negative correlation between holding time and *P. falciparum* ELISA absorbance, suggesting that detectable circumsporozoite protein and perhaps the number of sporozoites may decrease with time in the vector. In western Kenya, an area where *Anopheles* populations often have extremely high malaria infection rates, *Plasmodium* infections did not reduce vector survivorship.

KEY WORDS Insecta, *Anopheles* spp., *Plasmodium falciparum*, survival rate

THE EFFECT OF MALARIA PARASITES on the survival of their *Anopheles* hosts is potentially a limiting factor in malaria transmission. Decreased longevity of infected vectors would reduce the proportion of *Anopheles* that become infective and therefore their capacity to transmit malaria.

There have been no direct assessments of the effects of malaria parasites on the survivorship of naturally infected *Anopheles*. Studies on the effect of *Plasmodium* parasites on mosquito survival all have been done on mosquitoes either infected under laboratory conditions or fed on malaria patients. These experimental studies have indicated that *Plasmodium* either has no detrimental effect on the survivorship of mosquitoes (Wenyon 1926, King 1929, De Buck 1936, Freier & Friedman 1987) or that the parasite reduces the lifespan of infected mosquitoes (Buxton 1935; Gad et al. 1979; Klein et al. 1982, 1986). Klein et al. (1986) observed that the survival rate for *Anopheles dirus* Peyton & Harrison infected with *Plasmodium cynomolgi* Mayer was correlated inversely with the number of oocysts on the midgut. However, this effect was seen only when mosquitoes contained 10 or more oocysts.

In nature, it is difficult to measure how malaria infections affect vector survival. This is due to the logistical difficulties of finding and determining the infection status and age of dead mosquitoes. However, the effect of *Plasmodium* infection on the survivorship of vector populations should be examined for field vector populations, because laboratory induced infections may not closely simulate the natural situation. For example, "high" (>41) oocyst infections (Klein et al. 1986) are rarely observed in naturally infected Afrotropical *Anopheles* and >74% usually harbor <9 oocysts (Pringle 1966).

In our study, we examined the effect of malaria parasites on the survival rate and longevity of wild-caught, naturally infected *Anopheles*. By allowing mosquitoes sampled from wild populations to die in cages and then using an enzyme-linked immunosorbent assay (ELISA) for *P. falciparum* Welch, we compared the survivorship of infected versus uninfected *Anopheles*. We also examined the association between holding time and infection rates, the association between holding time and ELISA absorbance, and compared two methods of determining daily survival rates.

Materials and Methods

***Anopheles* Collection and Survival Experiments.** From November 1987 to March 1988, wild

[1] Current address: Department of Immunology and Infectious Diseases, The Johns Hopkins University, School of Hygiene and Public Health, 615 North Wolfe Street, Baltimore, Md. 21205.

Downloaded from http://jme.oxfordjournals.org/ at University of Iowa Libraries/Serials Acquisitions on June 7, 2016

An. gambiae Giles sensu lato and *An. funestus* Giles females were collected weekly by aspiration inside houses in Kisian, a locality 10 km west of Kisumu, western Kenya. Malaria transmission dynamics have been studied previously in this area (Fontaine et al. 1978, J. C. Beier et al. 1987). Females, collected from about 50 houses, were placed into screened 450-ml cartons and transported to Kisumu for processing. In the laboratory, mosquitoes were identified to species. Three to five replicates of 25 females of each species, irrespective of their trophic status, were established each week in screened 3.75-liter cartons. Mosquitoes were provided with 5% sucrose on cotton that was changed daily. Each cage was covered with a black cloth that was wetted twice daily. The cages were placed in a room (4 by 3 by 3 m) at ambient temperature and relative humidity in Kisumu. A hygrothermograph in the room showed that daily minimum and maximum temperatures throughout the study averaged 22.6 ± 0.9°C and 28.7 ± 1.5°C. Daily mean minimum and maximum relative humidities were 57.1 ± 8.5% and 74.3 ± 8.7%. Dead females were removed from each cage and counted daily until all females died.

Processing Dead Mosquitoes for ELISA. Dead females were placed individually into 1.5-ml polypropylene vials, held in a dessicator for 2 d, and then stored at room temperature. Later, the dry females (n = 2,349) were cut transversely at the thorax between the first and third pairs of legs and the anterior portion was ground in 50 μl blocking buffer with Nonidet P-40; 200 μl blocking buffer was then added, bringing the final volume to 250 μl (J. C. Beier et al. 1987). Samples were stored at −70°C and 50-μl aliquots were tested for *P. falciparum* circumsporozoite (CS) protein by an enzyme-linked immunosorbent assay (ELISA) using the 2A10 monoclonal antibody (Wirtz et al. 1987b). Of the 2,349 mosquitoes, 1,080 also were tested for CS proteins of *P. ovale* Stephens using the 110-54.3 monoclonal antibody and *P. malariae* Laveran using the 109-17.94 monoclonal antibody (M. S. Beier et al. 1988, Collins et al. 1988). Samples were considered positive if ELISA absorbance values exceeded twice the mean of four negative controls (J. C. Beier et al. 1988).

Determination of Parity Rates. Each week >30 (range, 32–95) *An. gambiae* s.l. and at least 25 (range, 25–48) *An. funestus* females captured resting inside houses in Kisian, 1 d earlier than those that were held in cages, were dissected to determine parity using the ovarian tracheation technique and by observing ovarian development (World Health Organization 1975).

Determination of Daily Survival Rates. Two methods for determining the daily survival rate (P) were used. The daily survival rate was calculated as the antilog of the slope (b) of the regression equation for the $\ln(y + 1)$ of the number alive per day (y) for each replicate of caged mosquitoes (52 for *An. gambiae* s.l. and 50 for *An. funestus*). The daily survival rate for each week was computed as the mean for all cohorts in that week. For field populations, survival rates were determined from the proportion parous (Davidson 1954), using the equation $P = (M)^{1/gc}$ where P = the survival rate, gc = the duration of the gonotrophic cycle, and M = the proportion parous. For both species, gc was considered to be 2 d (Davidson & Draper 1953).

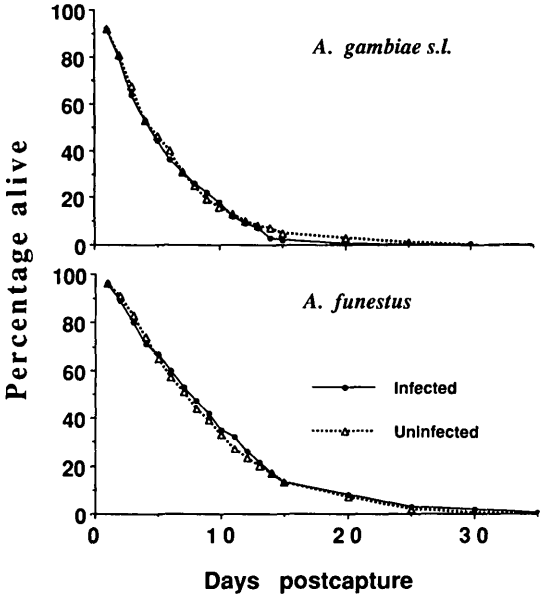


Fig. 1. Survivorship curves for uninfected and *P. falciparum* infected wild-caught *An. gambiae* s.l. and *An. funestus*. The curves show the percentage alive each day for the first 15 d after capture and at 5-d intervals thereafter. Note: females scored as uninfected include those that were gut infected but died before circumsporozoite protein was detectable by ELISA in the head–thorax region.

Results

Survivorship of Infected vs Uninfected Mosquitoes. There were no significant differences in the proportion surviving daily, between *P. falciparum* infected and uninfected *An. gambiae* s.l. (χ^2 = 0.084; df = 17; $P > 0.05$) or *An. funestus* (χ^2 = 0.037; df = 17; $P > 0.05$) (Fig. 1).

The number of days to 50% survival (P_{50}) also did not vary between infected and uninfected mosquitoes. For *An. gambiae* s.l., P_{50} occurred between days 4 and 5 and for *An. funestus* between days 7 and 8. The longest lived infected *An. gambiae* s.l. female died 22 d after capture, whereas the longest lived uninfected one survived 29 d after capture. The longest surviving infected *An. funestus* lived 3 d longer than its uninfected counterpart, which died 31 d after capture.

Mortality Rates. Caged *An. gambiae* s.l. had significantly higher daily mortality rates ($\bar{x}$ = 17.5%;

Downloaded from http://jme.oxfordjournals.org/ at University of Iowa Libraries/Serials Acquisitions on June 7, 2016

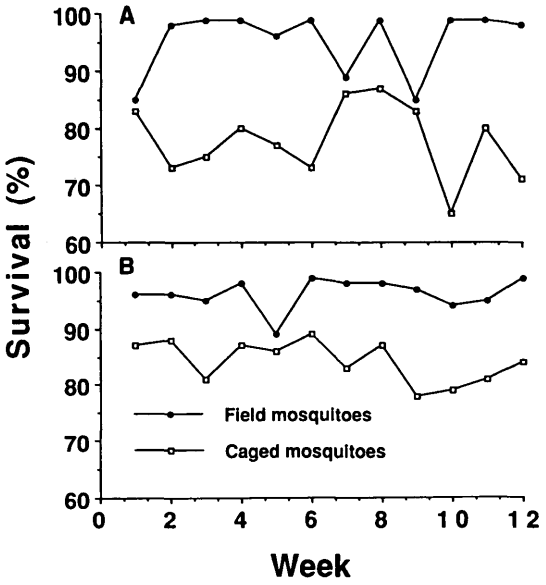


Fig. 2. Survival rates per week for wild-caught *An. gambiae* s.l. (A) and *An. funestus* (B) determined from the parity rates of dissected mosquitoes and by regression for caged mosquitoes.

n = 1,221) than *An. funestus* ($\bar{x}$ = 13.2%; n = 1,128) (F = 14.039; df = 1, 34; P = 0.0007).

Comparison of Survival Rates. Survival rates of field mosquitoes, which were determined by parity rates, were significantly higher than those determined by regression for caged *Anopheles* (Fig. 2).

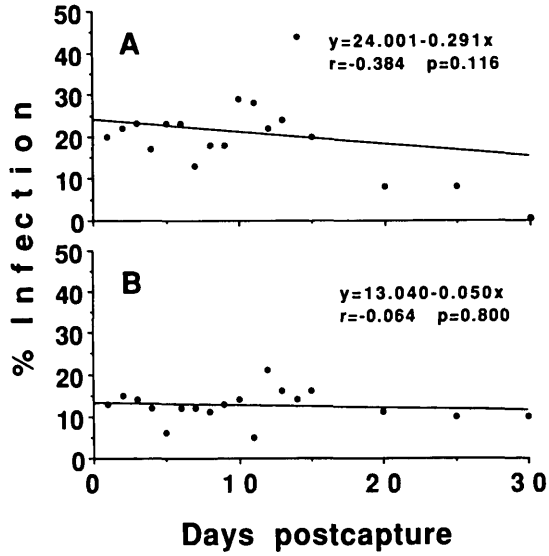


Fig. 3. *P. falciparum* ELISA infection rates per day (for the first 15 d and at 5-d intervals thereafter) after capture for caged naturally infected *An. gambiae* s.l. (A) and *An. funestus* (B). Regression lines are fitted; correlation coefficients (r) and probability values (P) are indicated.

This was true for *An. gambiae* s.l. (F = 49.69; df = 1, 22; P = 0.0001) and *An. funestus* (F = 118.69; df = 1, 22; P = 0.0001) and was consistent over the 12-wk observation period.

***Plasmodium falciparum* Infection Rates.** Daily infection rates which were higher for *An. gambiae* s.l. ($\bar{x}$ = 19.8%; n = 1,221) than for *An. funestus* ($\bar{x}$ = 11.9%; n = 1,128) did not change significantly after capture for *An. gambiae* s.l. (r = −0.384, df = 17, P = 0.116) or *An. funestus* (r = −0.064, df = 17, P = 0.800) (Fig. 3). *An. gambiae* s.l. weekly infection rates were significantly higher than those of *An. funestus* (F = 6.57, df = 1, 22, P = 0.018).

***Plasmodium malariae* and *P. ovale* Infection Rates.** The proportion of mosquitoes infected with *P. malariae* or *P. ovale* (0.7%, n = 1,080) was negligible. Therefore only *P. falciparum* infections were used in the analysis of the effect of *Plasmodium* on vector survivorship.

Correlation Between Absorbance and Holding Time. Regression of *P. falciparum* ELISA absorbance on day of death showed that there was a significant negative correlation for *An. gambiae* s.l. (r = −0.19, df = 244, P = 0.003) but not for *An. funestus* (r = −0.01, df = 138, P = 0.381).

Discussion

Natural infections of *P. falciparum* were not detrimental to the survivorship or longevity of field-collected *An. gambiae* s.l. and *An. funestus*. Infected mosquitoes did not die earlier or at a higher rate than did uninfected ones. Survivorship curves for infected and uninfected mosquitoes were nearly identical. Infection intensity in Afrotropical *Anopheles* may not be sufficient to affect mosquito survival, because most *An. gambiae* s.l. and *An. funestus* contain <10 oocysts (Pringle 1966, J. C. Beier et al. 1987). The experimental pathology described for heavily infected (>70 oocysts) *An. dirus* (Klein et al. 1986) has not been observed in dissections of >3,000 naturally infected *An. gambiae* s.l. and *An. funestus* in western Kenya (J.C.B., unpublished data). In Africa, *P. falciparum* infections probably have no detrimental effect on the survival rates of infected mosquitoes.

Davidson & Draper (1953), experimenting with wild-caught *An. gambiae* and *An. funestus* in Tanzania, observed that the former had lower mortality rates than *An. funestus*. In contrast, *An. gambiae* s.l. in this study had higher daily mortality rates than *An. funestus*. *An. funestus* also had a longer postcapture lifespan than did *An. gambiae* s.l.

Survival rates determined from the proportion of parous mosquitoes were much higher than those determined by regression. Gillies & Wilkes (1965) also observed that survival rates estimated from the proportion parous are high. However, the survival rates we obtained for caged wild-caught mosquitoes were only approximations for field popula-

Downloaded from http://jme.oxfordjournals.org/ at University of Iowa Libraries/Serials Acquisitions on June 7, 2016

tions, because the caged mosquitoes were exposed to natural environmental factors for only part of their lifespan, and gravid females were never allowed to oviposit or refeed.

Sporozoite rates increase naturally over time in caged, wild-caught *An. gambiae* and *An. funestus* because of sporogonic development and the migration of sporozoites to the salivary glands, and observations on delayed sporozoite rates have been used in estimating *Anopheles* survival rates (Davidson & Draper 1953). In our study, ELISA infection rates did not increase as a function of time but remained stable over the 30-d period. In western Kenya, ELISA infection rates for *An. gambiae* s.l. and *An. funestus* overestimate the true sporozoite rate by about 45%, because CS protein shed by oocysts and sporozoites (Verhave et al. 1988) is disseminated throughout the hemocoel. Sporozoites fail to appear in the salivary glands in about 10% of the infected *Anopheles*, even if they are held for 10 d to permit further sporogonic development (J.C.B., unpublished data). The progressive increase in infectivity, which is measured easily by dissection, therefore cannot be evaluated using ELISA techniques that do not differentiate between infected and infective mosquitoes.

Infected *An. gambiae* tended to lose ELISA detectable CS protein as a function of holding time. This could indicate a progressive decrease in sporozoites, as has been observed in *Aedes aegypti* L. infected with *P. gallinaceum* Brumpt (Porter et al. 1954), because ELISA absorbance is correlated positively with levels of CS protein and the number of sporozoites (Baker et al. 1987, Wirtz et al. 1987a). This observation should be tested further using quantitation techniques for salivary gland sporozoites (J. C. Beier et al. 1989). A decrease in sporozoite load over time would be significant epidemiologically in terms of vector potential.

Acknowledgment

We are grateful to M. Mugambi, former Director of the Kenya Medical Research Institute (KEMRI) for facilitating this research. We thank D. Koech, former Director, Biomedical Sciences Research Centre, KEMRI, and C. R. Roberts, Commander, U.S. Army Medical Research Unit, Kenya, for support and encouragement. We thank C. Oyaro, M. Omondi, L. Madahana, and other staff members of the Walter Reed Project for technical assistance. We also thank R. Copeland and C. R. Roberts for critically reviewing the manuscript. The views of the authors do not purport to reflect the position of the U.S. Department of the Army, the U.S. Department of Defense, or the Government of Kenya.

References Cited

Baker, E. Z., J. C. Beier, S. R. Meek & R. A. Wirtz. 1987. Detection and quantification of *Plasmodium falciparum* and *P. vivax* infections in Thai-Kampuchean *Anopheles* (Diptera: Culicidae) by enzyme-linked immunosorbent assay. J. Med. Entomol. 24: 534–541.

Beier, J. C., P. V. Perkins, R. A. Wirtz, R. E. Whitmire, M. Mugambi & W. T. Hockmeyer. 1987. Field evaluation of an enzyme-linked immunosorbent assay (ELISA) for *Plasmodium falciparum* sporozoite detection in anopheline mosquitoes from Kenya. Am. J. Trop. Med. Hyg. 36: 459–468.

Beier, J. C., C. M. Asiago, F. K. Onyango & J. K. Koros. 1988. Elisa cut-off method affects malaria sporozoite rate determination in wild Afrotropical *Anopheles*. Med. Vet. Entomol. 2: 259–264.

Beier, J. C., C. N. Oster, J. K. Koros, F. K. Onyango, A. K. Githeko, E. Rowton, D. K. Koech & C. R. Roberts. 1989. Effect of human circumsporozoite antibodies in *Plasmodium*-infected *Anopheles* (Diptera: Culicidae). J. Med. Entomol. 26(6): 547–553.

Beier, M. S., I. K. Schwartz, J. C. Beier, P. V. Perkins, F. Onyango, G. H. Campbell, P. M. Andrysiak & A. D. Brandling-Bennett. 1988. Identification of malaria species by ELISA in sporozoite and oocyst infected *Anopheles* from western Kenya. Am. J. Trop. Med. Hyg. 38: 323–327.

Buxton, P. A. 1935. The effect of *Proteosoma* upon the survival of *Culex*. Parasitology 27: 547–50.

Collins, F. H., P. M. Procell, G. H. Campbell & W. E. Collins. 1988. Monoclonal antibody-based enzyme-linked immunosorbent assay (ELISA) for detection of *Plasmodium malariae* sporozoites in mosquitoes. Am. J. Trop. Med. Hyg. 38: 87–91.

Davidson, G. 1954. Estimation of the survival rate of anopheline mosquitoes in nature. Nature (Lond.) 174: 792–793.

Davidson, G. & C. C. Draper. 1953. Field studies of some of the basic factors concerned in the transmission of malaria. Trans. R. Soc. Trop. Med. Hyg. 47: 522–535.

De Buck, A. 1936. Some results of six years' mosquito infection work. Am. J. Hyg. 24: 1–18.

Fontaine, R. E., J. H. Pull, D. Payne, G. D. Pradhan, G. P. Joshi, J. A. Pearson, M. K. Thymakis & M. E. Ramos Camacho. 1978. Evaluation of fenitrothion for the control of malaria. Bull. W.H.O. 56: 445–452.

Freier, J. E. & S. Friedman. 1987. Effect of *Plasmodium gallinaceum* infection on the mortality and body weight of *Aedes aegypti* (Diptera: Culicidae). J. Med. Entomol. 24: 6–10.

Gad, A. M., W. A. Maier & G. Piekarski. 1979. Pathology of *Anopheles stephensi* after infection with *Plasmodium berghei berghei*. I. Mortality rate. Z. Parasitenkd. 60: 249–261.

Gillies, M. T. & T. J. Wilkes. 1965. A study of the age composition of populations of *Anopheles gambiae* Giles and *A. funestus* Giles in north-eastern Tanzania. Bull. Entomol. Res. 56: 237–262.

King, W. V. 1929. On the development of malaria parasites in the mosquito. Am. J. Hyg. 10: 560–564.

Klein, T. A., B. A. Harrison, R. G. Andre, R. E. Whitmire & I. Inlao. 1982. Detrimental effects of *Plasmodium cynomolgi* infections on the longevity of *Anopheles dirus*. Mosq. News 42: 265–271.

Klein, T. A., B. A. Harrison, J. S. Grove, S. V. Dixon & R. G. Andre. 1986. Correlation of the survival rates of *Anopheles dirus* A (Diptera: Culicidae) with different infection densities of *Plasmodium cynomolgi*. Bull. W.H.O. 64: 901–907.

Porter, R. J., R. L. Laird & E. M. Dusseau. 1954. Studies on malarial sporozoites. II. Effect of age and

Downloaded from http://jme.oxfordjournals.org/ at University of Iowa Libraries/Serials Acquisitions on June 7, 2016

dosage of sporozoites on their infectiousness. Exp. Parasitol. 3: 267–274.

Pringle, G. 1966. A quantitative study of naturally acquired malaria infections in *Anopheles gambiae* and *Anopheles funestus* in a highly malarious area of East Africa. Trans. R. Soc. Trop. Med. Hyg. 60: 626–632.

Verhave, J. P., A. D. E. M. Leeuwenberg, T. Ponnudurai, J. H. E. T. Meuwissen & J. A. M. Van Druten. 1988. The biotin–streptavidin system in a two site ELISA for the detection of plasmodial sporozoite antigen in mosquitoes. Parasite Immunol. (Oxf.) 10: 17–31.

Wenyon, C. M. 1926. Protozoology: a manual for medical men, veterinarians, and zoologists, vol. 2. Bailiere, Tindall & Cox, London.

Wirtz, R. A., T. R. Burkot, P. M. Graves & R. G. Andre. 1987a. Field evaluation of enzyme-linked immunosorbent assays for *Plasmodium falciparum* and *Plasmodium vivax* sporozoites in mosquitoes (Diptera: Culicidae) from Papua New Guinea. J. Med. Entomol. 24: 433–437.

Wirtz, R. A., F. Zavala, Y. Charoenvit, G. H. Campbell, T. R. Burkot, I. Schneider, K. M. Esser, R. L. Beaudoin & R. G. Andre. 1987b. Comparative testing of *Plasmodium falciparum* sporozoite monoclonal antibodies for ELISA development. Bull. W.H.O. 65: 39–45.

World Health Organization. 1975. Manual on practical entomology in malaria. Part II. Methods and techniques. World Health Organization Offset Publication 13, Geneva, Switzerland.

Received for publication 21 November 1988; accepted 28 July 1989.

Downloaded from http://jme.oxfordjournals.org/ at University of Iowa Libraries/Serials Acquisitions on June 7, 2016